News-Driven Risk Budgeting

From financial news to portfolio weights

nabia — `python/` project

Results as of 2026-10-23

Abstract

This document describes the pipeline of the `python/` folder, step by step and with its mathematics. Financial news is collected into a SQLite database and scored by a language model on 27 economic factors. The portfolio's assets are mapped to those factors by estimating, from price data, how each asset moves with each factor. The factor sentiment, carried through those exposures, scales each asset's risk budget, and the nabia risk budgeting optimizer (C++) turns the budgets into long-only weights. Every step is point in time: at a date t it only uses information available before t. The database currently holds 17,757 articles (2020-01-01 to 2026-10-27), of which 16 have been scored so far.

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1 Overview

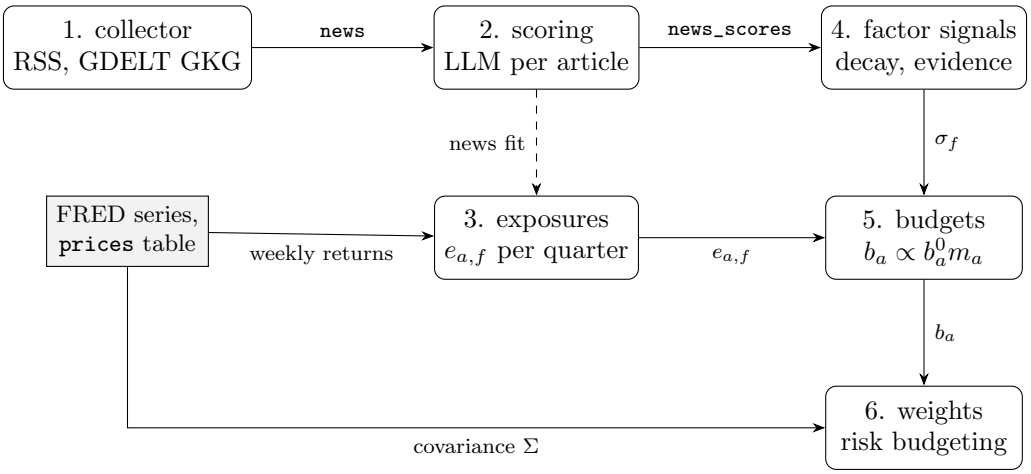


Figure 1: The six steps and the database tables between them.

1. **Collection** (`collector/`): RSS feeds (today's items) and GDELT GKG files (history) are stored in the `news` table, with duplicates removed.
2. **Scoring** (`scoring/`): one language-model call per article gives a sentiment $s \in [-1, 1]$ and a confidence $c \in [0, 1]$ for each factor the article concerns (`news_scores`).
3. **Exposures** (`exposures/`): every quarter, a rolling regression of each asset's weekly returns on orthogonalized factor proxies (and on the factors' news) gives the signed exposure $e_{a,f} \in [-1, 1]$ (`asset_exposures`).
4. **Factor signals** (`budgets/compute.py`): the scores are aggregated per factor into a decayed, evidence-weighted signal $\sigma_f(t) \in [-1, 1]$.
5. **Risk budgets**: the asset score $S_a(t) = \sum_f e_{a,f}\sigma_f(t) / \sum_f |e_{a,f}|$ scales the base budget, $b_a \propto b_a^0 (1 + \kappa S_a)$ (`risk_budgets`).
6. **Weights** (`budgets/weights.py`): the long-only weights whose risk contributions equal the budgets, from the C++ optimizer of `nabia_python_bridge` (`portfolio_weights`).

All settings are in `config/config.json`; the factors are in `config/factors.json`. Section 11 shows how to run each step.

2 Data layer

All data is in `data/market.db` (SQLite), read and written through the `datalayer` package: `NewsDB` for the news tables, `MarketDB` for the market tables. Times are UTC ISO 8601 strings, so they sort

and compare as text.

table	one row per	content
`prices`	symbol × day	daily OHLC and adjusted close of the assets and price proxies (pre-existing)
`assets`	symbol	name, asset class, currency (pre-existing; referenced by `prices`)
`news`	article	url, headline, body, source, `published_at`, `received_at`, `dedup_key`
`news_scoring`	article × model × prompt	relevant flag, scoring time
`news_scores`	article × model × prompt × factor	sentiment s, confidence c
`series`	series × day	downloaded FRED series (key `fred:<id>`)
`asset_exposures`	quarter end × asset × factor	$e_{a,f}$, raw beta, t-statistic, stability, method
`risk_budgets`	rebalance × asset	base budget, score S_a, multiplier m_a, budget b_a
`portfolio_weights`	rebalance × asset	budget, weight w_a, risk contribution, portfolio risk

3 News collection

Sources.

- **RSS and Atom feeds** (`collect.feeds`: MarketWatch, CNBC, Financial Times, Investing.com, OilPrice, Federal Reserve, ECB, Bank of England). Feeds carry no history, so they only give today's items; `received_at` is the download time.
- **GDELT Global Knowledge Graph.** Every 15 minutes GDELT publishes a file listing the web articles it has processed, with their page titles. For each weekday since `collect.since`, two 15-minute slices (14:30 and 20:30 UTC) are read, the titles matching a finance and macro vocabulary are kept, well-known financial outlets are ranked first, and at most `gkg_per_day` = 10 articles are stored. `received_at` is the GKG timestamp: when GDELT processed the article, a point-in-time stamp. GKG gives titles only, so the body is the headline.

Each run continues GKG from the day after the last GKG day stored. `--since` collects from a given day, and `--import` loads a `news.json` file of the former project.

Deduplication. With $\nu(h)$ the headline lower-cased and reduced to its alphanumeric words, an article is stored only if both its URL and its key

$$k = \nu(\text{headline}) \,\|\, \text{day}(\text{received_at})$$

are new (`UNIQUE` constraints and `INSERT OR IGNORE`). The same story syndicated by several outlets on the same day is therefore stored once.

4 Economic factors

`config/factors.json` lists 27 factors. The sign convention is the same for all of them:

> $s > 0$ means *the factor itself goes up*: more growth, more inflation, tighter policy, wider spreads, more stress, more conflict, a stronger currency, whether or not that is good news for markets.

This makes the sign independent of the asset. Whether a move of the factor is good or bad for an asset is learned from data in the exposure step (Section 6). Each factor has a `group` and a `region` tag, used for shrinkage and fallbacks. There is no tree: the drivers cut across asset classes (the Fed moves equities, bonds, gold and the dollar at once), so a hierarchy would force a single parent per node and make features collinear. Table 1 lists them, with the method that produced their current exposures.

Table 1: The factors, their tags and the method of their latest exposures (`proxy`: market proxy; `group`: borrowed from the group; `none`: no information yet).

factor	group	region	method	sentiment > 0 means
GROWTH_US	growth	US	proxy	stronger US activity, jobs, consumer, housing
GROWTH_EUROPE	growth	EU	proxy	stronger euro-area and UK activity
GROWTH_CHINA	growth	CN	proxy	stronger Chinese activity, stimulus, property stabilizing
GROWTH_EM	growth	EM	group	stronger emerging economies (ex China), capital inflows
TECH_AI	corporate	GLOBAL	proxy	stronger AI capex, semiconductor and cloud demand
EARNINGS	corporate	GLOBAL	group	earnings or guidance above expectations, margins up
POLICY_FED	policy	US	proxy	hawkish Fed: hikes, fewer cuts, balance-sheet tightening
POLICY_ECB	policy	EU	group	hawkish ECB: hikes, fewer cuts
POLICY_BOJ	policy	JP	group	BoJ tightening: hikes, less bond buying
POLICY_PBOC	policy	CN	group	PBoC tightening: less easing, higher rates or reserve ratios
POLICY_OTHER_DM	policy	GLOBAL	group	hawkish BoE, SNB, BoC, RBA and others
FISCAL_US	fiscal	US	proxy	wider deficits, more Treasury issuance, debt-ceiling stress
SOVEREIGN_EUROPE	fiscal	EU	group	budget or debt stress in euro-area countries, wider spreads
ENERGY_SUPPLY	commodity	GLOBAL	proxy	tighter oil and gas supply: OPEC+ cuts, outages, sanctions, inventory draws
INDUSTRIAL_METALS	commodity	GLOBAL	proxy	tighter copper, aluminium, iron-ore balance
AGRICULTURE	commodity	GLOBAL	proxy	tighter grain and soft-commodity supply: weather, export bans
INFLATION_US	inflation	US	proxy	higher or upside-surprising US prices and wages
INFLATION_EUROPE	inflation	EU	group	higher or upside-surprising euro-area and UK prices
CREDIT_STRESS	financial	GLOBAL	proxy	wider credit spreads, defaults, downgrades
BANKING_STRESS	financial	GLOBAL	proxy	bank troubles, funding or liquidity squeezes
RISK_APPETITE	financial	GLOBAL	proxy	investors adding risk: inflows, leverage, speculation
USD	fx	US	proxy	stronger dollar against the other major currencies
JPY	fx	JP	proxy	stronger yen: repatriation, carry unwind, intervention
CNY	fx	CN	proxy	stronger yuan, firmer PBoC fixing
GOLD_DEMAND	commodity	GLOBAL	proxy	central-bank buying, ETF inflows, safe-haven demand for gold
GEOPOLITICS	event	GLOBAL	none	more armed conflict, sanctions, political crises
TRADE_POLICY	event	GLOBAL	none	more tariffs, export controls, trade escalation

5 Scoring with a language model

5.1 The call

Each article gets one call. The *system prompt* holds the instructions, the sign rule, the scale and the factor list (`id | name | positive meaning`); it is identical for every article, so a local

model reuses its processed prefix and an API can cache it. The *user prompt* holds only the article: timestamp, source, headline and body (truncated to 4000 characters). The answer is constrained to the JSON schema

$$\{\ \texttt{relevant}: \text{bool},\ \ \texttt{scores}: [\,\{\texttt{factor} \in \mathcal{F},\ \texttt{sentiment},\ \texttt{confidence}\}\,]\ \}$$

through the provider's structured output: Ollama's `format`, OpenAI's `json_schema` (strict) or an Anthropic tool whose input is the schema. Temperature is 0.

The scale given to the model is ± 1 for a major, unambiguous move, ± 0.5 for a clear moderate one, ± 0.2 for a mild one and 0 for neutral or in-line news. The rules: use only the article; score the news against the expectations it states; reported price moves are reactions, not information; usually one to three factors; lower confidence for rumours, opinion and headline-only items.

5.2 Validation

For an answer with entries (f_j, s_j, c_j), unknown factors are dropped, one entry per factor is kept, and values are clipped:

$$s \leftarrow \mathrm{clip}(s, -1, 1), \qquad c \leftarrow \mathrm{clip}(c, 0, 1).$$

The article is *relevant* if the model says so and at least one valid score remains. A malformed answer or a network error is retried up to `llm.max_retries` times; an article still failing is skipped and retried on the next run.

5.3 Versions and resumption

The prompt version is $v = \mathrm{SHA256}(\text{system prompt} \,\|\, \text{user template})_{1..12}$. Since the system prompt contains the factor list, editing either the prompt or the factors changes v. A scoring is stored per (article, model, v), and only articles without a scoring for the current model and version are sent, so an interrupted run resumes where it stopped and a new version rescores everything, keeping the old scores.

5.4 Providers and throughput

`llm.provider` selects a local Ollama model (`gemma4:26b`, the default; about 5 s per article), the OpenAI Chat Completions API or the Anthropic Messages API, with the key read from the environment variable named in the configuration. `llm.workers` requests run in parallel threads (database writes stay in the main thread); HTTP 429 and 5xx answers are retried after the server's `Retry-After`, and `llm.requests_per_minute` spaces the request starts across workers. `--dry-run` reports the number of articles and their prompt tokens before an API run.

6 Asset exposures

The exposure $e_{a,f}$ answers: *when factor f moves up, how does asset a move?* It is estimated from prices, which have a long history, rather than from the news, which is short.

6.1 Weekly returns

Every daily series X is sampled at the last observation of each week (Friday), X_τ, and transformed into a weekly change:

$$\Delta_\tau X = \begin{cases} \ln X_\tau - \ln X_{\tau-1} & \text{prices, indices, FX, VIX (}\texttt{logret}\text{),} \\ X_\tau - X_{\tau-1} & \text{yields, spreads, stress indices (}\texttt{diff}\text{).} \end{cases}$$

Weekly returns avoid the bias of daily returns between markets closing at different hours (Tokyo, Hong Kong, Europe, New York). The proxy of factor f is a weighted sum of terms $(X_j, \text{transform}_j, \omega_j)$ from `factors.json`:

$$p_{f,\tau} = \sum_j \omega_j \, \Delta_\tau X_j.$$

factor	proxy p_f	source
GROWTH_US	$\Delta \ln$ Russell 2000	`prices`
GROWTH_EUROPE	$\Delta \ln$ Euro Stoxx 50	`prices`
GROWTH_CHINA	$\Delta \ln$ Hang Seng	`prices`
TECH_AI	$\Delta \ln$ Nasdaq 100 $-\ \Delta \ln$ S&P 500	`prices`
POLICY_FED	Δ 2-year Treasury yield (`DGS2`)	FRED
FISCAL_US	Δ 10-year term premium (`THREEFYTP10`)	FRED
ENERGY_SUPPLY	$\Delta \ln$ Brent future	`prices`
INDUSTRIAL_METALS	$\Delta \ln$ copper future	`prices`
AGRICULTURE	$\frac{1}{3}(\Delta \ln$ corn + soy + wheat futures)	`prices`
INFLATION_US	Δ 10-year breakeven inflation (`T10YIE`)	FRED
CREDIT_STRESS	Δ Baa $-$ 10-year spread (`BAA10Y`)	FRED
BANKING_STRESS	Δ St. Louis Fed financial stress index (`STLFSI4`)	FRED
RISK_APPETITE	$-\Delta \ln$ VIX (`VIXCLS`)	FRED
USD	$\Delta \ln$ broad dollar index (`DTWEXBGS`)	FRED
JPY	$-\Delta \ln$ yen per dollar (`DEXJPUS`)	FRED
CNY	$-\Delta \ln$ yuan per dollar (`DEXCHUS`)	FRED
GOLD_DEMAND	$\Delta \ln$ gold future	`prices`

The HY option-adjusted spread is no longer distributed by FRED beyond three years, hence the Baa spread. Ten factors have no market proxy (the non-Fed central banks, European sovereign risk and inflation, emerging-market growth, earnings, geopolitics, trade policy).

6.2 Orthogonalization

At each quarter end q, take the last $T = 156$ weeks (`window_weeks`). Proxies with less than 90% of the weeks are left out, and weeks with any missing value are dropped. Let $P \in \mathbb{R}^{T \times K}$ hold the proxies in the order of `factors.json`, $X = P - \bar{P}$ its centred columns and $y \in \mathbb{R}^T$ the asset's centred weekly log returns. Proxies are correlated (equities and the 2-year yield, Brent and growth), so they are made orthogonal in that order, with the QR decomposition

$$X = QR, \qquad Q^\top Q = I_K, \quad R \text{ upper triangular.}$$

Proposition 1. *The k-th column of QR_{kk} (with R_{kk} the diagonal entry) is the residual z_k of the regression of x_k on $x_1, \ldots, x_{k-1}$: proxy k net of the proxies before it. The residuals are mutually orthogonal and $\|z_k\| = |R_{kk}|$.*

Proof. $x_k = \sum_{j \le k} R_{jk} q_j$, and $q_1, \ldots, q_{k-1}$ span the same space as $x_1, \ldots, x_{k-1}$. The projection of x_k on that space is $\sum_{j<k} R_{jk} q_j$, so the residual is $R_{kk} q_k$, orthogonal to the earlier columns and to the other q_j. □

The order is an *identification assumption*: the variance two proxies share is attributed to the earlier one. The file puts equity growth first, so the 2-year yield net of equities is read as a monetary-policy shock; Brent net of growth as an energy-supply shock; breakevens net of growth and energy as inflation; gold last, net of the dollar and the Fed.

6.3 Exposures and t-statistics

Regressing y on the orthogonal residuals $z_1, \dots, z_K$ gives, column by column,

$$\beta_k = \frac{z_k^\top y}{z_k^\top z_k} = \frac{q_k^\top y}{R_{kk}}, \qquad \hat{\varepsilon} = y - QQ^\top y, \qquad \hat{\sigma}^2 = \frac{\|\hat{\varepsilon}\|^2}{T - K - 1}.$$

The exposure is the standardized coefficient and the t-statistic uses $\mathrm{se}(\beta_k) = \hat{\sigma}/\|z_k\|$:

$$\boxed{e_{a,k}^{\mathrm{raw}} = \beta_k \frac{\|z_k\|}{\|y\|} = \mathrm{sign}(R_{kk})\, \frac{q_k^\top y}{\|y\|}, \qquad t_{a,k} = \frac{\beta_k}{\mathrm{se}(\beta_k)} = \mathrm{sign}(R_{kk})\, \frac{q_k^\top y}{\hat{\sigma}}.}$$

$e_{a,k}^{\mathrm{raw}}$ is the correlation of the asset with the factor's shock, in $[-1, 1]$, and since the q_k are orthonormal,

$$\sum_k \left(e_{a,k}^{\mathrm{raw}}\right)^2 = \frac{\|QQ^\top y\|^2}{\|y\|^2} = R_a^2,$$

so the squared exposures split the share of the asset's weekly variance that the factors explain.

6.4 News fit

For a factor with news, its weekly sentiment is the confidence-weighted mean of the scores of the articles received that week, an article scored by several models or prompt versions counting once with its mean score:

$$\bar{s}_{f,\tau} = \frac{\sum_{i \in \tau} c_{i,f}\, s_{i,f}}{\sum_{i \in \tau} c_{i,f}}.$$

If at least $n_{\min} = 52$ weeks of the window have news (`news_min_weeks`), the news fit is the correlation r between $\bar{s}_{f,\cdot}$ and the asset's returns over those n weeks, with $t = r\sqrt{(n-2)/(1-r^2)}$. It fills the factors without a proxy and, with `news_override`, replaces the proxy fit when $|t| \geq t_{\min}$: news captures what a proxy cannot, such as a Fed shock on equities, which the ordering above assigns to growth.

6.5 Selection, shrinkage and fallback

For asset a at quarter q, with fits $(e_{a,f}^{\mathrm{raw}}, t_{a,f})$ by method $\mu \in \{\text{proxy}, \text{news}\}$:

1. **Significance and stability.** With $\mathcal{H}$ the last 8 fits of the same pair and method (`stability_windows`), the stability is

   $$\pi_{a,f} = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \mathbf{1}\left[\,\mathrm{sign}\, e_h = \mathrm{sign}\, e_{a,f}^{\mathrm{raw}}\right],$$

   and the exposure is kept only if it is significant and stable:

   $$\tilde{e}_{a,f} = e_{a,f}^{\mathrm{raw}}\, \mathbf{1}\left[\,|t_{a,f}| \geq t_{\min} \;\wedge\; \pi_{a,f} \geq \pi_{\min}\right], \qquad t_{\min} = 2,\ \pi_{\min} = 0.75.$$

2. **Shrinkage toward the group.** With $G(f)$ the fitted factors of f's group and $\bar{e}_{a,G} = \frac{1}{|G|}\sum_{g \in G} \tilde{e}_{a,g}$,

   $$e_{a,f} = (1 - \lambda)\, \tilde{e}_{a,f} + \lambda\, \bar{e}_{a,G(f)}, \qquad \lambda = 0.25 \ (\texttt{group_shrink}).$$

3. **Fallback.** A factor without a fit borrows $e_{a,f} = \gamma\, \bar{e}_{a,G(f)}$ with $\gamma = 0.5$ (`group_fallback`); if its group has no fit, $e_{a,f} = 0$.

Each (q, a, f) is stored with e, e^{raw}, t, π and the method. `MarketDB.exposures(t)` returns the map of the latest quarter end before t, so a backtest only uses exposures that were known.

6.6 Results

Figure 2 shows the map at 2026-09-30 (of 56 quarter ends since 2012). The structure is economically sensible: bonds load on the Fed in order of duration sensitivity to short rates (2-year -0.94, 10-year -0.86, 30-year -0.73) and the 30-year on fiscal supply (-0.57); equities on growth and the AI cycle; each commodity on its own supply factor; gold also negatively on the dollar and the Fed. The non-Fed central banks show half of the Fed's exposures (fallback) until their news fit becomes available. Figure 3 shows how some exposures moved over time.

7 Factor signals

At a rebalance date t, the signal of factor f uses the articles received strictly before day t. With d_i the receipt day of article i, its age $\Delta_i = t - d_i \geq 1$ days, the half-life H (`half_life_days` $= 5$) and $\delta = 2^{-1/H}$, the decayed sums are

$$N_f(t) = \sum_{i:\, d_i < t} \delta^{\,\Delta_i - 1}\, c_{i,f}\, s_{i,f}, \qquad D_f(t) = \sum_{i:\, d_i < t} \delta^{\,\Delta_i - 1}\, c_{i,f}.$$

They are computed recursively over calendar days, $N(t) = \delta N(t-1) + \sum_{i:d_i = t-1} c_i s_i$ (and the same for D). The level is the decayed, confidence-weighted mean sentiment, and the signal shrinks it by the amount of evidence:

$$\ell_f(t) = \frac{N_f(t)}{D_f(t)} \in [-1, 1], \qquad \boxed{\sigma_f(t) = \ell_f(t)\, \frac{D_f(t)}{D_f(t) + \eta}} \qquad \eta = 3 \text{ (`evidence`)},$$

with $\sigma_f = 0$ when $D_f = 0$. D_f counts the recent articles, weighted by confidence and age: a factor with one confident article from yesterday has $D_f \approx 1$ and keeps a quarter of its level; with twelve, $D_f \approx 12$ and it keeps 80%. The sign of σ_f follows the factor convention: $\sigma_f > 0$ means the news says the factor is going up.

8 Risk budgets

8.1 Asset score

With $e_{a,f}$ the latest exposure map estimated before t,

$$\boxed{S_a(t) = \frac{\sum_f e_{a,f}\, \sigma_f(t)}{\sum_f |e_{a,f}|}} \in [-1, 1],$$

with $S_a = 0$ for an asset without exposures. It is the exposure-weighted average of the factor signals, each signed by how the asset responds: hawkish Fed news ($\sigma_{\text{FED}} > 0$) times the 2-year note's $e = -0.94$ lowers the note's score; energy-supply news lifts Brent's. $S_a > 0$ means the news is bullish for the asset. The contribution of factor f, $e_{a,f}\sigma_f / \sum_g |e_{a,g}|$, is printed by the command line, so every budget change can be traced to its news.

8.2 Budgets

With base budgets b_a^0 (`budgets.base`, equal when null), strength $\kappa = 0.5$ and bounds $[m_{\min}, m_{\max}] = [0.5, 1.5]$:

$$m_a(t) = \text{clip}\left(1 + \kappa\, S_a(t),\ m_{\min},\ m_{\max}\right), \qquad \boxed{b_a(t) = \frac{b_a^0\, m_a(t)}{\sum_j b_j^0\, m_j(t)}}.$$

Budgets stay positive (the optimizer needs $b_a > 0$) and sum to one. With $|S_a| \leq 1$ and $\kappa = 0.5$, an asset's budget can at most be multiplied by 1.5 or halved relative to the others. Budgets are computed every Friday (`budgets.frequency`) since `budgets.start`.

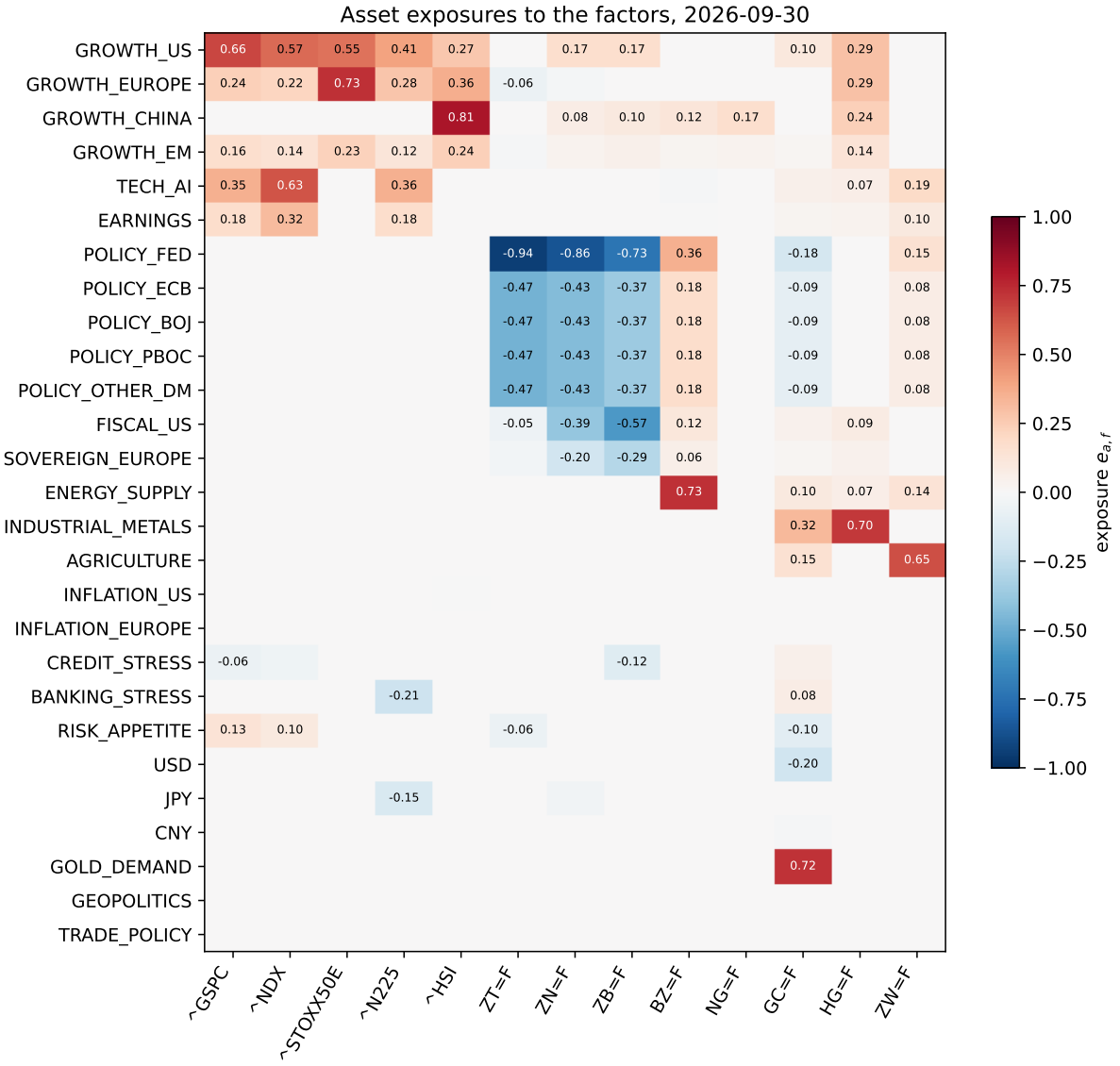


Figure 2: Exposures $e_{a,f}$ at 2026-09-30. Blank cells are zero: not significant, not stable, or no information.

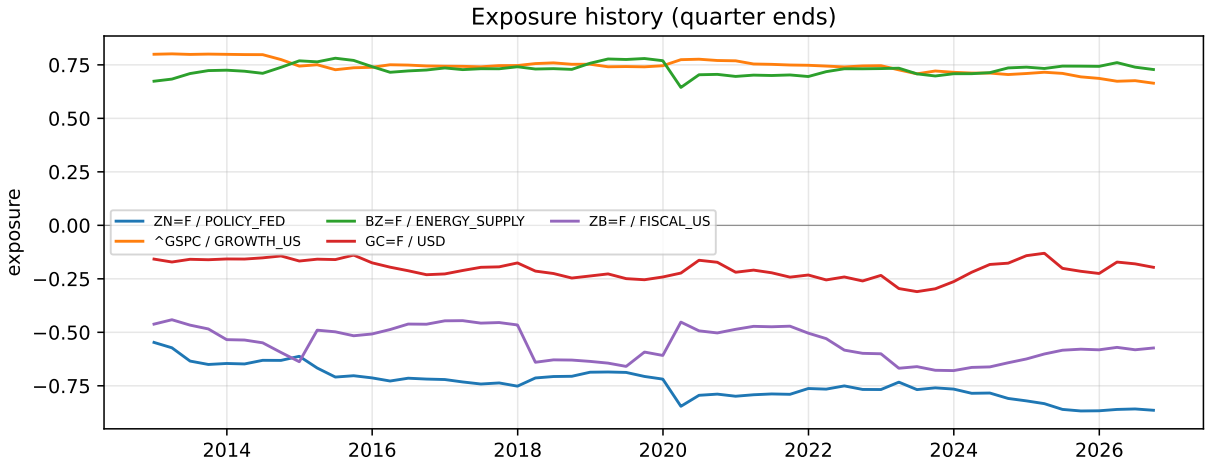


Figure 3: Exposure history of selected asset–factor pairs (zero when the tests fail).

9 Weights by risk budgeting

9.1 Covariance

At rebalance t, Σ is the sample covariance of the assets' weekly simple returns over the last 104 weeks up to t (`cov_window_weeks`), annualized ($\times 52$) and shrunk toward its diagonal:

$$\Sigma = (1-\rho)\,\hat{\Sigma} + \rho\,\operatorname{diag}(\hat{\Sigma}), \qquad \rho = 0.1 \ (\texttt{shrinkage}).$$

The shrinkage keeps Σ well conditioned with 13 assets and 104 observations; at least 80% of the window must have complete data.

9.2 The risk budgeting problem

For a risk measure $\mathcal{R}(w)$ that is convex and positively homogeneous of degree one, Euler's theorem gives $\mathcal{R}(w) = \sum_a w_a\,\partial_a\mathcal{R}(w)$, and the *risk contribution* of asset a is $RC_a = w_a\,\partial_a\mathcal{R}(w)$. With the volatility (`risk_measure` = `Volatility`),

$$\mathcal{R}(w) = \sqrt{w^\top \Sigma w}, \qquad RC_a = \frac{w_a(\Sigma w)_a}{\sqrt{w^\top \Sigma w}}.$$

The risk budgeting portfolio is the long-only, fully invested w whose relative contributions equal the budgets:

$$w_a > 0, \quad \sum_a w_a = 1, \quad \frac{RC_a(w)}{\mathcal{R}(w)} = b_a \quad \forall a.$$

Proposition 2 (Roncalli; Spinu)**.** *Let $y^\star$ solve the strictly convex problem*

$$\min_{y>0} \; F(y) = \mathcal{R}(y) - \sum_a b_a \ln y_a.$$

Then $w = y^\star / \sum_a y_a^\star$ is the unique risk budgeting portfolio.

Proof. The first-order condition is $\partial_a\mathcal{R}(y) = b_a/y_a$, i.e. $y_a\partial_a\mathcal{R}(y) = b_a$: the risk contributions of y are the budgets, and $\mathcal{R}(y) = \sum_a b_a = 1$. Since $\partial_a\mathcal{R}$ is homogeneous of degree zero, $w = y/\sum y$ has $RC_a(w)/\mathcal{R}(w) = RC_a(y)/\mathcal{R}(y) = b_a$. F is strictly convex (convex $\mathcal{R}$ plus a strictly convex barrier) and tends to $+\infty$ at the boundary and at infinity, so the minimizer exists and is unique. □

The C++ implementation (`nabia::risk::RiskBudgeting`) minimizes F with L-BFGS; the log barrier keeps $y > 0$ (the line search rejects steps leaving the domain). It stops when $\max_a |RC_a/\mathcal{R} - b_a|$ is below the tolerance (10^{-10} here). Each rebalance starts from the previous weights, which are close to the solution. The Gaussian Value-at-Risk and Expected Shortfall can be used instead (`risk_measure`):

$$\mathrm{VaR}_\alpha(w) = z_\alpha \sqrt{w^\top \Sigma w}, \qquad \mathrm{ES}_\alpha(w) = \frac{\phi(z_\alpha)}{1-\alpha} \sqrt{w^\top \Sigma w}, \qquad z_\alpha = \Phi^{-1}(\alpha),$$

which, without expected returns, are multiples of the volatility and give the same weights.

9.3 Results

All 356 weekly rebalances converged, with every risk contribution equal to its budget. Since only 16 articles are scored, all received after the last rebalance, the budgets are still equal and the portfolio is the equal risk contribution portfolio. Its annualized volatility is 2.78% (Table 2). The 2-year note receives most of the capital because its volatility is a fraction of the others': equal risk needs inverse-volatility-like weights. A portfolio with a volatility target would lever it up.

asset	budget b_a	weight w_a	risk contribution $RC_a/\mathcal{R}$
`^GSPC`	0.0769	0.0296	0.0769
`^NDX`	0.0769	0.0224	0.0769
`^STOXX50E`	0.0769	0.0321	0.0769
`^N225`	0.0769	0.0270	0.0769
`^HSI`	0.0769	0.0232	0.0769
`ZT=F`	0.0769	0.5837	0.0769
`ZN=F`	0.0769	0.1150	0.0769
`ZB=F`	0.0769	0.0589	0.0769
`BZ=F`	0.0769	0.0312	0.0769
`NG=F`	0.0769	0.0102	0.0769
`GC=F`	0.0769	0.0242	0.0769
`HG=F`	0.0769	0.0164	0.0769
`ZW=F`	0.0769	0.0263	0.0769

Table 2: Budgets, weights and risk contributions at 2026-10-23.

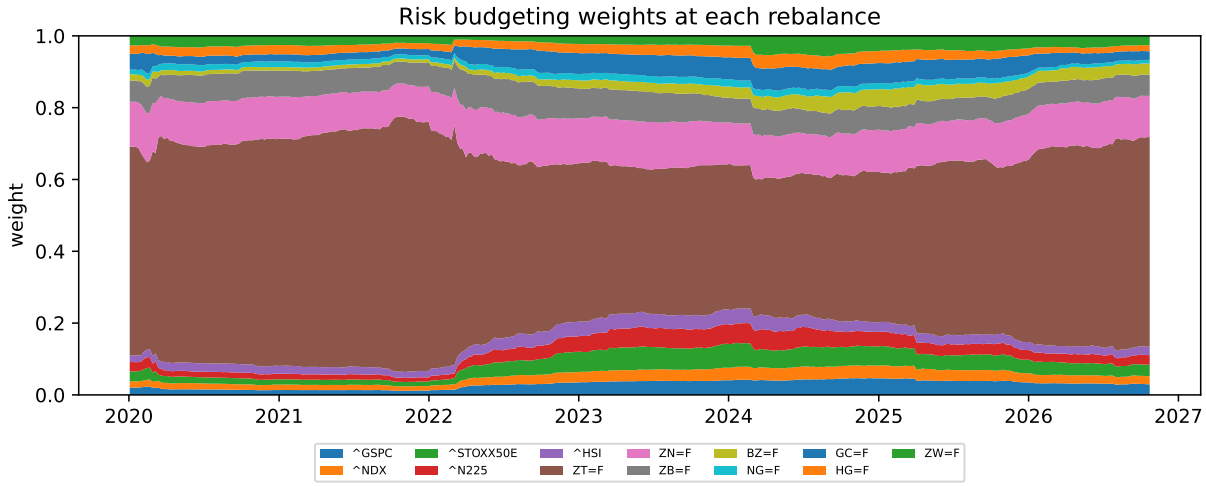


Figure 4: Weights at each weekly rebalance.

10 Point in time

Every step only uses what was known at the date it computes:

- `received_at` is the time an article became available (download time, GKG processing time), not the publication time an outlet may backdate.
- The factor signal at t uses articles received before day t; the budgets at t use the exposure map of the latest quarter end before t.
- Each exposure uses returns up to its quarter end only; its stability test uses earlier quarters only.
- The covariance at t uses returns up to t.

Two things are not point in time: the language model knows what happened after the articles it scores (the prompt forbids using that knowledge, but this cannot be guaranteed), and FRED series can be revised after their release.

11 Running the pipeline

From `python/`; the first run of `run.sh` creates `.venv` with numpy, pandas and the `nabia_python_bridge` wheel from `../dist`.

step	command (`py` = `.venv/bin/python`)
all steps	`./run.sh [config.json]`
1. collect	`py -m collector [--since D] [--until D] [--import news.json]`
2. score	`py -m scoring [--provider P] [--workers N] [--limit N] [--dry-run]`
3. exposures	`py -m exposures [--no-download]`
4–6. budgets, weights	`py -m budgets [--start D] [--end D]`
tests	`py -m unittest discover -s tests`
this document	`doc/build.sh`

Configuration. The main parameters of `config/config.json`:

section	parameters	values
`collect`	`since`	2020-01-01
	`gkg_per_day`, `gkg_slices`	10; 14:30 and 20:30 UTC
`llm`	`provider`, `workers`, `requests_per_minute`	ollama, 1, none
`exposures`	`window_weeks`, `t_min`	156, 2
	`stability`, `stability_windows`	0.75, 8
	`group_shrink`, `group_fallback`	0.25, 0.5
	`news_min_weeks`, `news_override`	52, true
`budgets`	`half_life_days`, `evidence`, `strength`	5, 3, 0.5
	`min_multiplier`, `max_multiplier`	0.5, 1.5
	`frequency`	weekly (Friday)
`weights`	`risk_measure`	Volatility
	`cov_window_weeks`, `shrinkage`	104, 0.1

12 Limitations and next steps

- **Scoring backlog.** 16 of 17,757 articles are scored. Until the backlog is scored, budgets stay at their base and the news fits of the exposures are inactive.
- **Identification by ordering.** Equity indices are the growth proxies, so equities get no Fed exposure from the proxies; the news fit restores it once enough Fed news is scored. Other orders are possible and change the map.

- **Headlines only.** GKG gives titles, a weaker signal than full text, and ten articles a day is a thin sample per factor.
- **Exposure is not prediction.** The exposures measure co-movement. Whether a factor's news *leads* returns is not tested here; a backtest of the budgets against equal risk contribution would measure it.
- **Futures** are continuous front-month series, so roll jumps add noise to their weekly returns.